

Analytics Studio

Product Requirements Document (PRD)

Version 3.0 (Final MVP)

Internship: ReadyNest – Week 6 Final Project

Deadline: Friday, 10:00 PM

Development Style: Fresh build, modular architecture, prompt-driven development using Antigravity with manual verification after every module.

Previous Internship Learning (Reference Only):

- Week 1 – Dataset Analysis & Reporting
- Week 2 – Customer Insights
- Week 3 – Google Maps Business Analysis
- Week 4 – Predictive Analytics
- Week 5 – End-to-End Data Pipeline

No code reuse.

Only concepts and learnings influence this project.

1. Vision

Analytics Studio transforms raw business data into actionable business decisions.

Rather than stopping at dashboards and machine learning predictions, the platform guides users through a complete decision-making workflow.

It answers three business questions in sequence:

What happened?

↓

Why did it happen?

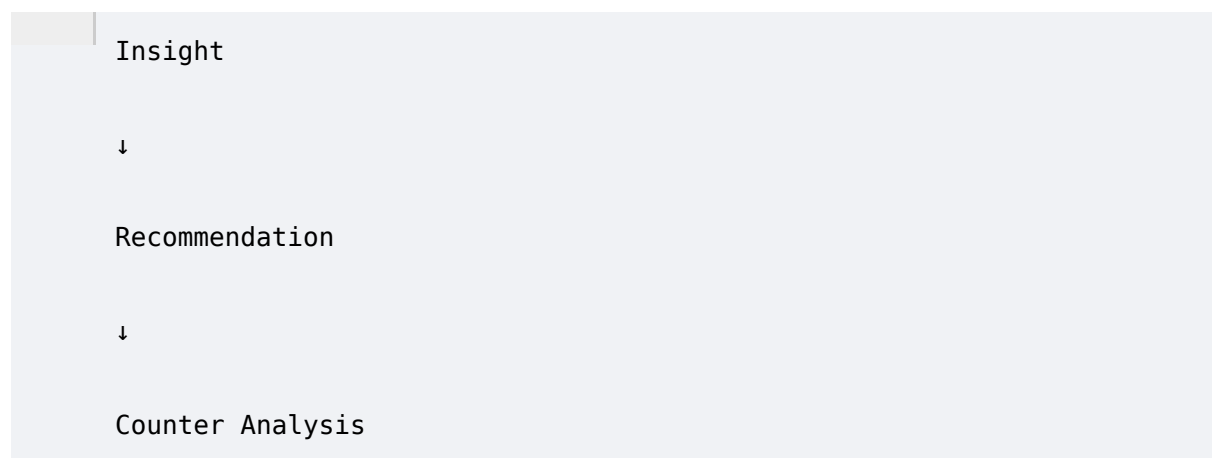
↓

What should I do next?

The platform is designed as a **Decision Intelligence Platform**, not merely an analytics dashboard.

2. Product Philosophy

Analytics Studio follows one reasoning framework throughout the application.



Every major output follows this pattern.

Instead of simply showing numbers, the platform explains findings, recommends actions, and highlights potential risks or alternatives.

3. Naming Convention (Locked)

The word **"AI"** will never appear as a feature prefix.

The intelligence is implementation detail, not branding.

Avoid	Use
AI Executive Report	Executive Report
AI Copilot	Executive Advisor
AI Decision Simulator	Decision Simulator
AI Reporting Suite	Reporting Suite

AI Dashboard Builder	Dashboard Builder
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4. Product Goals

Users should be able to:

- Upload datasets
- Assess data quality
- Explore data visually
- Generate predictive models
- Understand business insights
- Generate executive reports
- Simulate business decisions
- Explain charts
- Export deliverables

without leaving one application.

5. Target Users

- Business Owners
 - Students
 - Data Analysts
 - Researchers
 - Consultants
 - Small & Medium Enterprises (SMEs)
-

6. MVP Scope

Core Spine (Must Work End-to-End)

Dataset Module

- CSV Upload
- Dataset Preview
- Data Type Detection
- Missing Value Detection

- Duplicate Detection
 - Quality Score
-

Analytics Engine

- Summary Statistics
 - Correlation Analysis
 - Outlier Detection
 - Distribution Analysis
 - Interactive Charts
-

Prediction Engine

Automatic problem detection:

- Classification
- Regression

Outputs:

- Metrics
 - Predictions
 - Feature Importance
-

Insight Engine

Converts analytics into plain-English business findings.

Outputs structured insights instead of plain text.

Executive Report

Single PDF containing:

- Dataset Summary
 - Key Findings
 - Trends
 - Risks
 - Recommendations
 - Counter Analysis
-

Decision Simulator

Supports one business scenario for MVP.

Example:

Price Increase

↓

Expected Impact

↓

Confidence

↓

Assumptions

↓

Counter Analysis

↓

Recommendation

Executive Advisor

Context-aware explanation.

Example:

User clicks

Explain this chart

↓

Platform explains the visualization in business language.

This is **not** a chatbot.

7. Stretch Goals

Only after the complete MVP works.

- PowerPoint Export
- Second Decision Scenario
- Dashboard Builder (LLM-powered intent mapping)

Stretch goals will never block MVP completion.

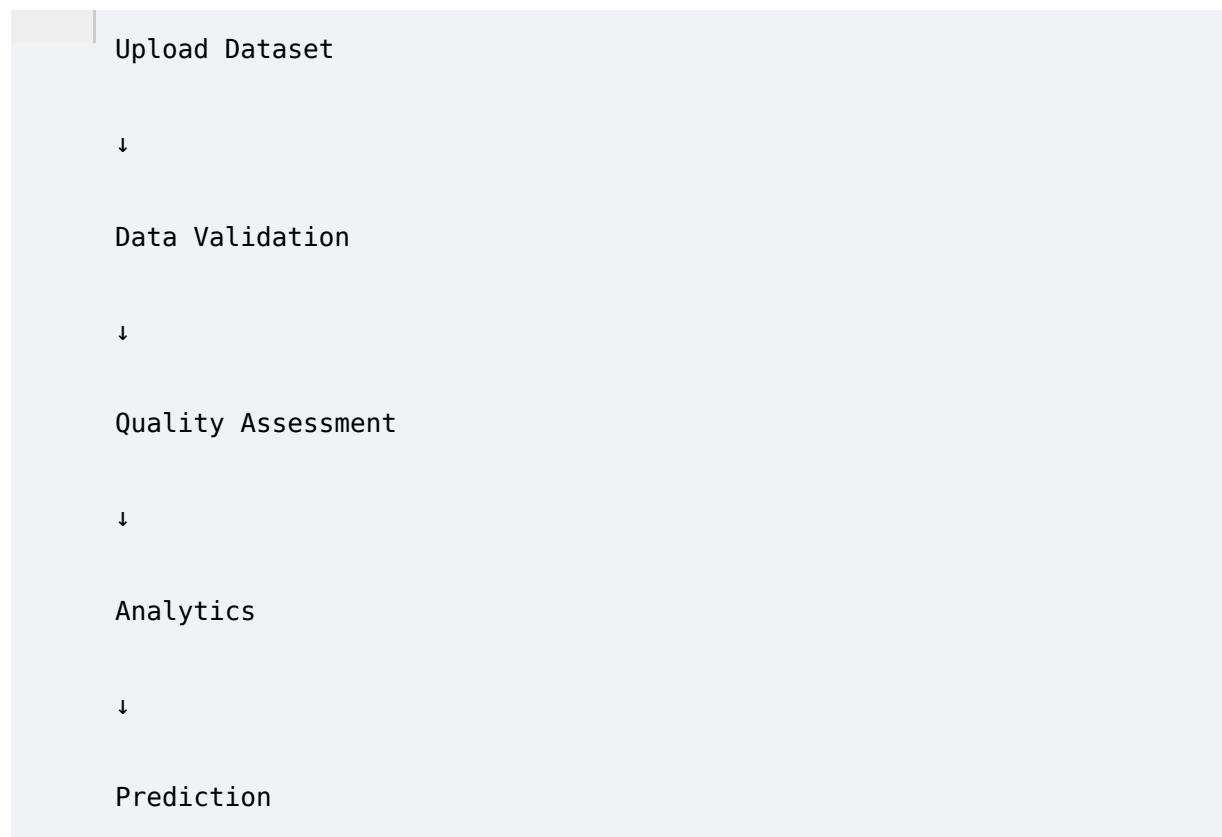
8. Features Explicitly Out of Scope

The following will **not** be implemented in the MVP.

- Authentication
- Multi-user support

- Database persistence
 - Team collaboration
 - Scheduling
 - Notifications
 - Forecasting
 - External API integrations
 - Open-ended conversational chatbot
 - Dynamic dashboard generation
 - Workspaces
 - Role management
-

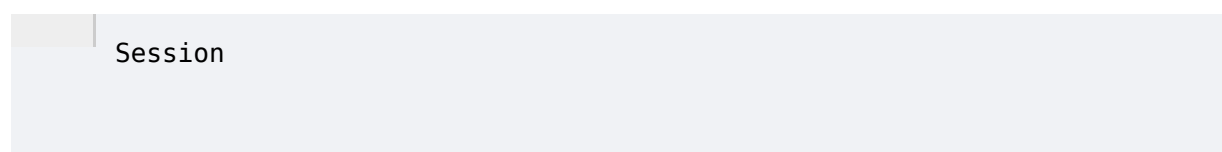
9. User Journey



10. System Architecture

Storage

Temporary in-memory storage only.



```
↓  
  
SessionState  
  
↓  
  
Dataset  
  
Analytics  
  
Prediction  
  
Insights  
  
Decision
```

No database.

No authentication.

Each uploaded dataset creates an isolated session.

11. Backend Architecture

```
backend/  
  
main.py  
  
models.py  
  
schemas.py  
  
storage.py  
  
modules/  
  
    dataset.py  
  
    analytics.py  
  
    prediction.py
```

12. Frontend Architecture

```
frontend/  
  
src/  
  
pages/  
  
Overview.tsx  
  
Analytics.tsx  
  
Predictions.tsx  
  
Report.tsx  
  
DecisionCenter.tsx  
  
components/
```

13. API Endpoints

```
POST /dataset/upload
```

↓

Dataset

```
GET /dataset/{id}/analytics
```

↓

Analytics

```
POST /dataset/{id}/predict
```

Body

```
{
```



```
target_column
```

```
}
```

```
↓
```

```
Prediction
```

```
GET /dataset/{id}/insights
```

```
↓
```

```
list[Insight]
```

```
POST /dataset/{id}/decision
```

```
{
```

```
scenario_name,
```

```
input
```

```
}
```

```
↓
```

```
Decision
```

```
GET /dataset/{id}/report.pdf
```

```
↓
```

```
Executive Report (PDF)
```

```
POST /dataset/{id}/advisor
```

```
{
```

```
chart_id
```

```
or
```

```
insight_id
```

```
}
```

```
↓
```

```
AdvisorResponse
```

14. Canonical Domain Objects

Dataset

```
Dataset
```

```
id
```

```
filename
```

```
rows
```

```
columns
```

```
schema
```

```
quality
```

Analytics

```
Analytics
```

```
summary
```

```
correlations
```

```
charts
```

```
outliers
```

Prediction

	Prediction
	problem_type
	metrics
	feature_importance
	predictions

Insight

	Insight
	id
	title
	description
	severity
	category
	evidence
	confidence

Decision

	Decision
	scenario_name
	expected_impact
	confidence
	assumptions

recommendation

counter_analysis

Report

Report

sections

generated_at

pdf_path

15. Module Contracts

Dataset Module

Input

CSV

Output

Dataset

Analytics Engine

Input

Dataset

Output

Analytics

Prediction Engine

Input

Dataset

Output

Prediction

Insight Engine

Input

Analytics

Prediction

Output

```
list[Insight]
```

Report Module

Input

Dataset

Analytics

Prediction

Insights

Decision

Output

Report

Decision Simulator

Input

Analytics

Prediction

Scenario Input

Output

Decision

Executive Advisor

Input

Chart

or

Insight

Output

```
AdvisorResponse
```

```
{
```

```
    explanation
```

```
}
```

16. Architecture Principles (Locked)

Rule 1

Every module returns structured objects.

Never formatted text.

Rule 2

Modules never communicate using raw dictionaries.

Only canonical objects.

Rule 3

Only the Dataset Module owns the DataFrame.

Every other module consumes structured objects.

Rule 4

Presentation layer performs zero calculations.

All business logic stays in backend modules.

Rule 5

LLM components never read raw CSV data.

They consume Analytics, Prediction, Insight, and Decision objects only.

Rule 6

Every endpoint returns typed Pydantic models.

No anonymous responses.

Rule 7

One source of truth.

Dashboard, Report, Decision Simulator, and Executive Advisor render the same underlying objects instead of recalculating results.

17. Technology Stack

Frontend

- React
 - TypeScript
 - Tailwind CSS
 - shadcn/ui
 - Recharts
-

Backend

- FastAPI
 - Pandas
 - Scikit-learn
 - Pydantic
-

Reporting

- ReportLab
 - python-pptx (*Stretch Goal*)
-

LLM

Provider-agnostic.

Used only for:

- Insight generation
 - Executive Advisor
 - Counter Analysis
 - Executive Report narrative
-

18. Development Timeline

Tuesday Evening – Wednesday

- Dataset Module
- Analytics Engine
- Prediction Engine

Verification:

Run multiple datasets.

Validate contracts.

Thursday

- Insight Engine
- Executive Report
- Executive Advisor

Verification:

Read reports manually.

Verify insight quality.

Friday (Morning / Afternoon)

- Decision Simulator
- Counter Analysis
- Dashboard integration

Verification:

Complete upload → report → decision flow.

Friday Evening

- Bug Fixes
- UI Polish
- Demo Rehearsal

No new features.

19. Success Criteria

The project is considered complete when a user can:

1. Upload a CSV dataset.
2. View cleaned analytics and visualizations.
3. Generate predictions.
4. Read structured business insights.
5. Download an Executive Report.
6. Simulate a business decision.
7. View recommendation with counter analysis.

8. Explain any chart using Executive Advisor.
without any manual intervention.

20. Unique Selling Proposition (USP)

Analytics Studio doesn't stop at analytics—it transforms raw business data into informed business decisions through structured insights, executive reporting, contextual explanations, and decision simulation.

21. Future Roadmap (Post-MVP)

- Authentication
 - Multi-user workspaces
 - Database persistence
 - Team collaboration
 - Dashboard Builder (fully natural language)
 - Additional Decision Simulator scenarios
 - Forecasting
 - Scheduled reports
 - External data connectors
 - Enterprise role management
 - Presentation themes & branding
-

Final Product Identity

Analytics Studio is **not** a dashboard.

It is **not** a machine learning application.

It is **not** a chatbot.

It is a **Decision Intelligence Platform** that takes users from:

Raw Data → Analytics → Insights → Recommendations → Counter Analysis → Business Decisions